

**Investigate the impact of Confounding Medical Interventions (CMI) in
electronic health records (EHR) data**

Statistical Science Honors Thesis

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Abstract

Clinical prediction models (CPMs) using electronic health records (EHRs) data capture patient outcomes inaccurately when medical intervention is present. We considered two intervention scenarios implemented in medical services and illustrated the factors that influence the performance of CPMs: the effectiveness of intervention, the prevalence of outcome, and the percentage of population receiving intervention. Then, we proposed two correction methods that improve CPM performance. Under a deterministic intervention scenario, stratified intervention approach is superior, especially when the outcome is rare. Under a probabilistic intervention scenario, the adjusted posterior risk is superior, especially when the outcome is common.

Thanks

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1. Introduction

Electronic health records (EHRs) data are widely used for the development of clinical prediction models (CPMs). The abundance of available data, the low cost of obtaining these data, and their representation of a real-world population, make them a highly appealing data source (Goldstein BA 2017). As EHR derived CPMs proliferate, health systems are increasingly implementing these models into clinical work-streams. These models are used in a variety of ways to inform and impact clinical care. For example, some models are used as strict decision-rules where any patient above a certain risk threshold receives a clinical intervention (Chen 2020). Others are used as triaging tools to inform patient risk, potentially – but not necessarily – directing patients to additional care (Coslovsky et al., 2015).

As CPMs find their way into clinical practice, it is important to develop systems and procedures to monitor their safety and effectiveness. The Food and Drug Administration (FDA) has developed draft guidance for how such tools should be considered (Food and Drug Administration 2022). In time-critical contexts, for example, CPM are only allowed to provide recommendations to healthcare professionals (HCP) to enhance or inform a health care decision but not replace or direct the HCP's judgment. Within Duke University Health System, a governance framework has been developed for evaluating and monitoring the performance of CPMs to ensure their adequate performance. A key part of that of that process is the establishment of an ongoing process to evaluate the performance of tools being used. The intention of such monitoring plans is to – including other factors – ensure that the model is performing as expected as clinical populations and workflows evolve (Bedoya et al., 2022).

While evaluating model performance using traditional metrics such as area under the receiver operating characteristic (AUROC) and calibration is seemingly straightforward, upon

reflection a clear paradox emerges. As mentioned, after implementing a CPM, medical professionals will use these models to inform clinical decision-making and allocation of interventions. Ideally, the intervention will prevent the outcome that the CPM is predicting. If the CPM is accurate, and the intervention is effective, those who are truly at risk of the adverse outcome will not actually have the event. This leads to an important challenge: how do we evaluate CPMs in the presence of interventions?

To evaluate the CPM appropriately, we are interested in what the causal inference refers to as the counterfactual (Judea 2000): the outcome that would have occurred without the intervention being deployed. Unfortunately, we do not observe this counterfactual. Current research on EHR data has investigated several sources of biases such as selection bias (Haneuse et al., 2016), informed presence bias (Goldstein et al., 2016), and confounding (Li et al., 2014), and proposed approaches to control these biases. However, little work has considered this potential bias which has been labeled Confounding by Medical Interventions (CMIs). Many prediction studies did not consider the influence of CMI on the true outcomes (Shavdia 2007). The little literature on this topic has focused on how we can learn new CPMs in the presence of CMI (Paxton 2013). Systematic bias emerges as interventions performed by caregivers affect the risk of outcome and mask the true condition of patients, making it difficult to distinguish those who were at risk but treated and those who were healthy but received unnecessary treatment. The existing suggestions to mitigate such bias include 1) using data with no CMI for model training or 2) treating all observations with CMI as definitive labels. They are either treated as positive due to trust in medical professionals' judgment or negative assuming the prognosis are too conservative. Although the researchers were able to improve the classification accuracy using the first approach, ignoring data with CMI during training reduced the generalizability of the model

to validation data. Assigning all data with CMI with definitive labels did not perform well enough. Alternatively, treating confounded data as unlabeled and inferring the true labels using transductive support vector machine (Joachims 1999, Paxton 2013) did not perform better than the approach that ignores any data with CMI.

In general, the literature is not well developed. This project seeks to study how CMI can impact our ability to evaluate CPMs and the strategies we can use to address these challenges. We first propose possible causal models. Next, we use simulation to both illustrate the problem and explore potential solutions. Then, we will evaluate these approaches in real-world healthcare data.

2. Naïve evaluation of Clinical Prediction Model in the presence of Confounding by Medical Intervention

2.1 Theoretical Model for Confounding by Medical Intervention

To illustrate how CMI arises, we first consider the key components of a medical intervention driven by a CPM, illustrating how it can generate CMI. The factors included in the causal models are the patient's:

- initial risk, as indicated by the CPM, reflecting the *true* probability of how likely that patients will have the outcome.
- intervention received, indicated as 1 = intervention, 0 = no intervention.
- a new (lower) risk, after the receipt of intervention. The change in risk is reflected by $\beta_{intervention}$.
- the observed outcome, indicated as 1 = sick, 0 = healthy.

Additionally, we consider how the underlying outcome prevalence impacts the system. As such, we envision a patient will receive an intervention depending on their initial risk. The effectiveness of treatment is determined by $\beta_{intervention}$. In general, the smaller $\beta_{intervention}$ (i.e., farther from 1) the greater the effectiveness. New risks (risk after intervention) are calculated for patients who should have had outcomes due to higher initial risk and also received intervention treatment. Depending on the treatment strength, the patient who would have had the outcome, may not have the outcome. As such, in the real world, only the initial risk and observed outcome are known to health professionals. As can be seen in Figure 1, the presence of intervention influences both the initial risk we observed as well as the risk after intervention, which determines the observed outcome.

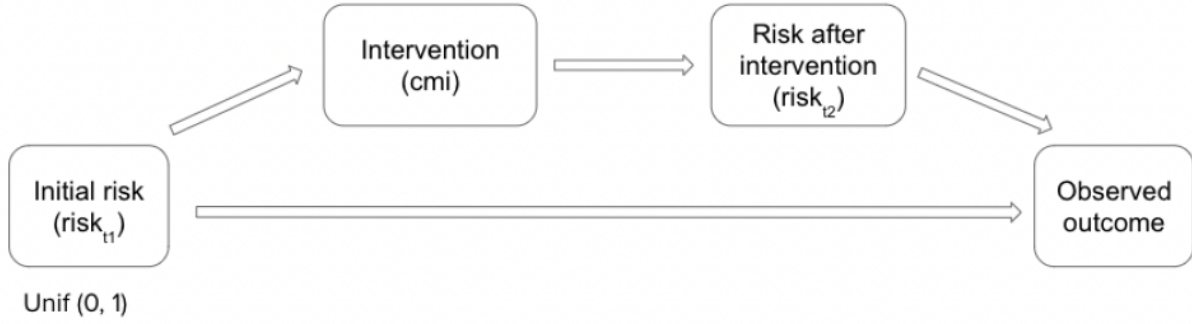


Figure 1: Causal diagram

With this causal diagram in mind, we next consider two processes under which interventions can be generated. The first process is based on a deterministic intervention. In this case, patients with risks above a predetermined threshold (indicated by parameter *cmi cutoff*) are eligible to receive the intervention. The second process is based on a triaging model, where patients with higher risk are more likely to receive an intervention but not guaranteed to. As we will illustrate below, these two processes have different implications for how we evaluate our CPM.

2.2 Simulation of Data Confounded by Medical Intervention

With these components in mind, we describe our simulation process:

1. $n = 20000$ observations are assigned an initial risk $risk_{t1}$ uniformly distributed from 0 to 1.
2. Observed outcome (Y) without intervention is determined by a binomial distribution as following: $x = prevalence + c \times risk_{t1}$, where c is a constant to control

AUROC at around 0.8 ($c = 5$). $p = \frac{e^x}{1+e^x}$, so that $Y \sim binomial(n, p)$.

3. In deterministic intervention, intervention (cmi) is present when a patient has $risk_{t1}$ higher than a set cmi_cutoff . In probabilistic intervention, intervention (cmi) is proportional to $risk_{t1}$. Define $x = prevalence + m \times risk_{t1}$, where m is a tuning parameter that controls the degree of connection between $risk_{t1}$ and intervention. The larger m is, the more likely an intervention is registered given the same change in $risk_{t1}$. $p = \frac{e^x}{1+e^x}$, so that $cmi \sim binomial(n, p)$.
4. $risk_{t2} = risk_{t1} \times \beta_{intervention}$ if a patient had an outcome and received treatment. Otherwise, $risk_{t2}$ remains the same as $risk_{t1}$.
5. For patients who had an outcome and received treatment, their observed outcome (with intervention) is re-generated in the same way in step 2, while the outcomes of the rest remain to be their outcomes without intervention.

We simulate three levels of outcome prevalence rate in the absence of intervention: high (~50% population have the outcome), moderate (~25% population have the outcome), and low (~6% population have the outcome). With the confounding intervention variable, we ideally want to achieve a prediction power as good as predicting outcomes directly from the known initial risk. These metrics are averaged over 100 simulations, with each simulation of 20000 observations.

We generate simulated EHR datasets with 20000 patients repeatedly. In each iteration, the data is split into training (80%) and testing (20%). A binary logistic regression model is fit on training data such that $logit(Y) = \beta_0 + \beta_r \times risk_{t1}$. The first evaluation metric is AUROC, indicating the model's performance at distinguishing between patients with and without outcomes. AUROC is calculated using the probability of the predicted outcome and true outcome

from the test set. The second evaluation metric is calibration slope, representing the degree of which model predictions are too high or too low compared to the true outcomes (Stevens and Poppe, 2019). To calculate the calibration slope, the true outcome from the test set is regressed on the linear combination of predicted outcome using a binary logistic regression model such that $\text{logit}(y) = \gamma_0 + \gamma_p \times \hat{y}$ where y is the true outcome and $\hat{y}$ is the linear predictor using the original coefficients β . γ_p is the calibration slope.

2.3 Results for evaluating clinical prediction model in the presence of intervention

We describe our results based on our two primary intervention scenarios: a deterministic intervention rule and a probabilistic intervention rule. Under a simulation framework, the expected model performance was an AUROC of 0.82 and a calibration slope of 1.0.

2.3.1 CMI under a Deterministic Intervention

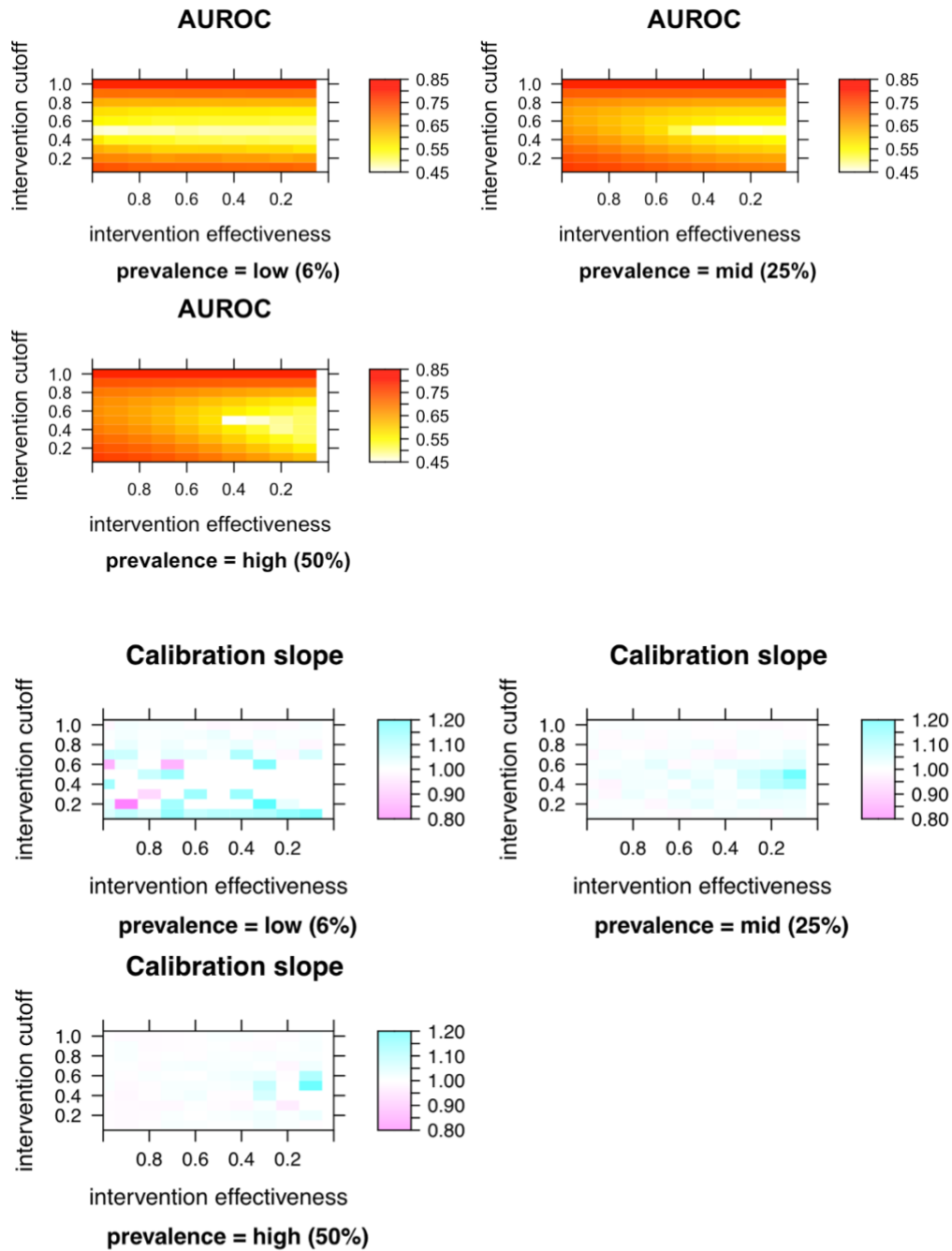


Figure 2: (a) AUROC decreases as intervention becomes more effective (b) More over- and under-estimation as intervention becomes more effective

The observed model performances across our simulations for a deterministic intervention are shown in Figure 2. Clear patterns emerge with respect to the estimated AUROC. For one, the observed model performance is highly influenced by the effectiveness of intervention ($\beta_{intervention}$) as well as the level for receipt of intervention (cmi_cutoff). For outcomes that are moderately or highly prevalent, less effective intervention is associated with higher AUROC hence better predictive performance. This is intuitive because the power of confounding decreases as the intervention becomes weaker, and the predicted outcomes are closer to the ones under direct causal relationship between initial risk and outcome. We also observe that CMI is worst for intervention rates closest to 50%, becoming less impactful as intervention rates approach 0 and 100% respectively. Because AUROC is a rank measure of the probability of a healthy person being ranked more highly than a sick person or vice versa, having half of the population ($cmi_cutoff = 0.5$) receive treatment presents the worst scenario where the most amount of people have the opportunity to be “re-shuffled”. With regard to prevalence, for rarer outcomes, the influence of intervention effectiveness is not obvious in Fig. 2, possibly due to the lack of outcome. Nonetheless, when the intervention is not effective at all, all models under 3 prevalence scenarios achieved the nearly ideal AUROC score controlled at the start of simulation.

Patterns are less clear for calibration slope. A calibration slope of 1 is a perfectly calibrated model (no under- or over- estimation). Less than 1 (pink) means overestimating for patients at high risk, while larger than 1 (blue) means underestimating for patients at low risk. Overall, more prevalent outcomes have better calibration. For the rarer outcomes with 6% occurrence rate, there is a trend of underestimating initial risk when most patients get

intervention ($cmi_cutoff = 0.1$). For more prevalent outcomes, the more effective intervention is, the worse the calibration is.

To further quantify the impact of 1) the criteria of intervention, 2) intervention effectiveness, and 3) prevalence on model performance, we regress $\beta_{intervention}$, cmi_cutoff , and prevalence on AUROC.

$$\begin{aligned} AUROC = & \beta_0 + \beta_1 \times cmi\ cutoff + \beta_2 \times cmi\ cutoff^2 \\ & + \beta_{intervention} \times intervention\ effectiveness + \beta_p \times prevalence \end{aligned}$$

We include a quadratic term for cmi-cutoff to reflect the non-linear relationship of the effect. The model output (Table 1) matches our previous observations. The less effective an intervention is, the higher AUROC is. Compared to low prevalence outcome (baseline), moderately and highly prevalent outcomes both have higher overall AUROC. For a scenario in which the intervention cutoff is at $risk_{t1} = 0.1$, requiring patients to have an initial risk 10% higher to receive intervention is associated with an AUROC change of -0.8718. Under intervention cutoff at $risk_{t1} = 0.5$, the AUROC change is -0.023. Under intervention cutoff at $risk_{t1} = 0.9$, the AUROC change is 0.8258. In other words, under strict intervention criteria, raising the bar for intervention is associated with improved model performance, but model performance becomes worse when the bar is raised under relaxed intervention criteria.

Table 1: Relationship between AUROC and predictors

	estimate	95% CI
Intercept	0.764	[0.746, 0.782]
cmi cutoff	-1.084	[-1.146, -1.021]
(cmi cutoff) ²	1.061	[1.006, 1.116]
intervention effectiveness	0.097	[0.083, 0.111]
prevalence (mid vs. low)	0.040	[0.03, 0.049]
prevalence (high vs. low)	0.038	[0.028, 0.047]

Table 2: Relationship between AUROC and predictors

	estimate	95% CI
Intercept	0.681	[0.655, 0.707]
intervention effectiveness	0.076	[0.037, 0.115]
prevalence (mid vs. low)	-0.018	[-0.042, 0.007]
prevalence (high vs. low)	-0.040	[-0.065, -0.016]

2.3.2 Model evaluation under a probabilistic intervention

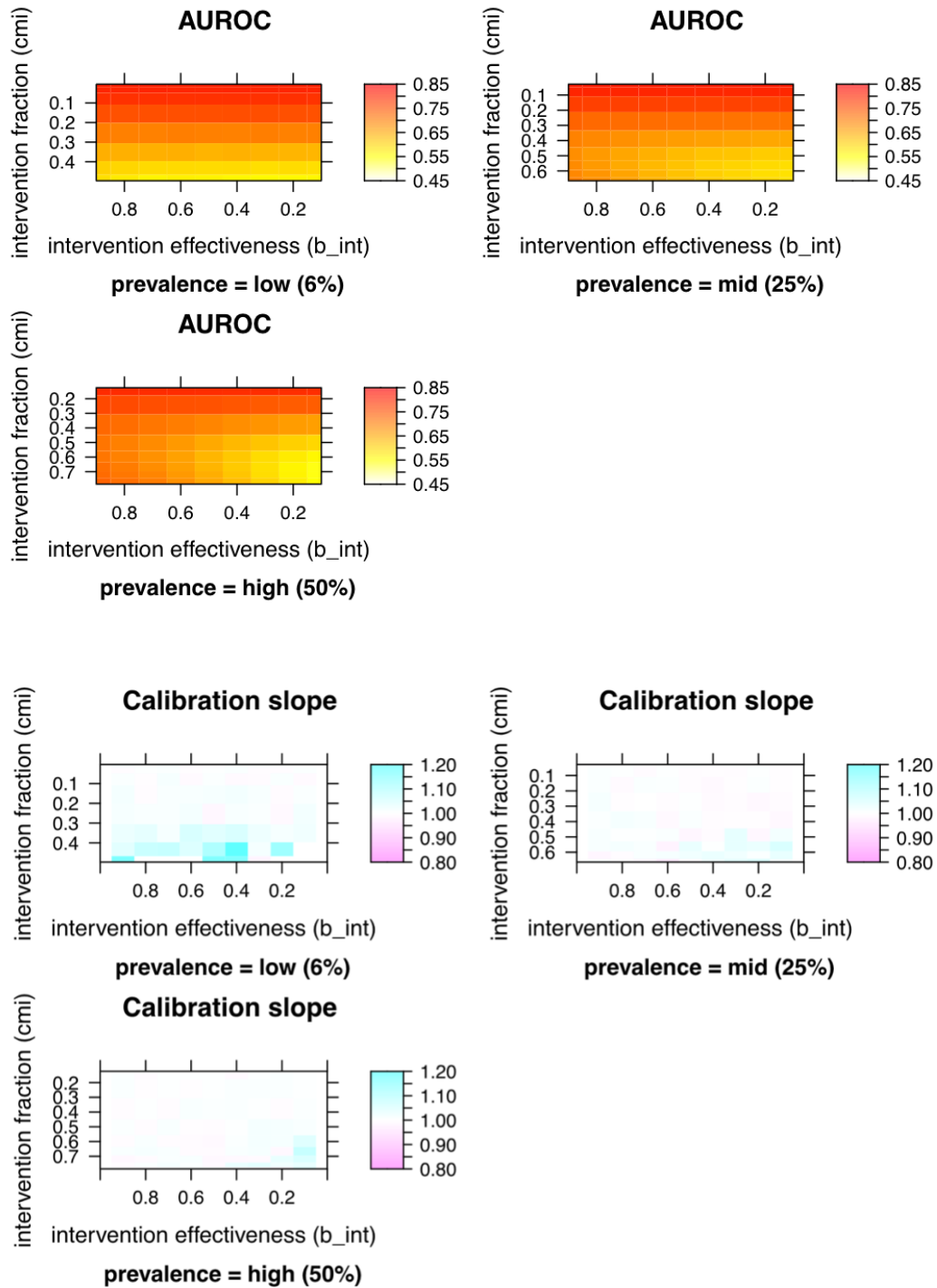


Figure 3: (a) AUROC decreases as intervention becomes more effective at mid/high prevalence(b)

Calibration is consistent around 1 indicating little over- or under- estimation

Under the probabilistic intervention scenario (Fig. 3), there are no strict criteria for intervention. Rather, patients at higher risk are more likely to receive the intervention. The results are similar to the first scenario. The model performance is highly influenced by the effectiveness of intervention $\beta_{intervention}$ as well as outcome prevalence. For outcomes that are moderately or highly prevalent, less effective intervention is associated with higher AUROC hence better predictive performance. For rarer outcomes, the influence of intervention effectiveness is not obvious. Different from the first scenario, the model achieves near-perfect AUROC only in the rare outcome case. In general, the model is well calibrated except that it underestimates for patients to be at low risk when the outcome is rare and fraction of patients receiving intervention is high. This underestimation reflects the challenge of predicting outcomes in the presence of CMI, as the model learns to label a large percentage of patients as having no outcome because their intervention has masked the true outcome that are supposed to happen.

To quantify the impact of 1) intervention effectiveness and 2) prevalence on model performance, we regress $\beta_{intervention}$ and prevalence on AUROC.

$$AUROC = \beta_0 + \beta_{intervention} \times \text{intervention effectiveness} + \beta_p \times \text{prevalence}$$

Consistent with the first scenario, the less effective an intervention is, the higher AUROC is. However, outcome prevalence has a different impact on AUROC from the first scenario. CPM for moderately and highly prevalent outcomes both have lower overall AUROC compared to rare outcomes (baseline).

In summary, Confounding by Medical Interventions (CMIs) is a challenge to the effective prediction of clinical models under many scenarios. First, the criteria of intervention directly affect the percentage of the population who receives treatment. The clinical model works best when this percentage is very large or very small in a deterministic intervention scenario.

Second, the more effective intervention is, the weaker the ability of our classifier is to distinguish between healthy and sick subjects. The prevalence of outcome also affects our overall confidence in predictions. After illustrating the research question by simulating health risks under two scenarios of intervention, we propose two potential solutions.

3. Evaluating predictive models with knowledge of confounding

After illustrating the problem generated by CMI, we next consider two potential solutions for correcting for it, when evaluating a model. We note that each approach requires knowledge that at an intervention was received in response to the CPM, something that is not always present in the EHR. The first approach evaluates the model within stratified samples while the second approach “corrects” the estimated risk based on the intervention receipt. For this part we focus on the impact of AUROC.

3.1 Methods for Correcting for CMI

Stratified intervention approach

We start by dividing our sample into two groups: intervened ($cmi = 1$) and non-intervened ($cmi = 0$). The model performance (AUROC) is calculated in each group by comparing the true outcome and $risk_{t_1}$. Then, we assigned a weight based on the number of observations in that group over the total observations. The final AUROC is the weighted average over two groups defined as:

Weighed AUROC

$$\begin{aligned} &= \text{percentage of observations in intervention group} \\ &\times \text{AUROC of intervention group} \\ &+ \text{percentage of observations in non-intervention group} \\ &\times \text{AUROC of non-intervention group} \end{aligned}$$

Adjusted posterior risk by estimating the effect of intervention

In this approach our goal is to adjust the estimated risk (from the CPM) for the fact that the individual received the intervention. We do this by first estimating the risk reduction due to intervention, via a log-binomial regression model, regressing the observed outcome on the knowledge of initial risk $risk_{t1}$ and cmi such that

$$\log(Y) = \beta_0 + \beta_r \times risk_{t1} + \beta_{cmi} \times cmi$$

The estimate of cmi (β_{cmi}) is considered to be a representation of the underlying effectiveness of treatment. Since this model is on the log scale, the exponentiation of this term can be interpreted as a relative risk (McNutt et al., 2003). Using this new estimate, we adjust the patients' initial risk if they have intervention ($cmi = 1$) by multiplying $risk_{t1}$ with $e^{\beta_{cmi}}$.

Then, we evaluate the new model using AUROC metric by comparing the true outcome and the adjusted risk, $risk_{t1} \times e^{\beta_{cmi}}$. Finally, we compare how close the model AUROC is to the ideal AUROC score controlled at the start of simulation for each of the approaches. The difference between model AUROC and ideal AUROC is defined as AUROC bias.

3.2 Simulation Framework

40000 observations are simulated based on the process specified in section 2.2, and model performance are averaged over 100 repetitions. Two scenarios of intervention are considered: deterministic intervention and probabilistic scenario. In each intervention scenario, we compare 2 approaches: stratified intervention approach and adjusted posterior risk by estimating the effect of intervention.

3.3 Results

3.3.1 Deterministic intervention results

The stratified intervention approach achieves a slightly better mitigation of AUROC bias at all levels of outcome prevalence (Fig. 4a). Both correction methods have a better effect in reducing the bias when an outcome occurs rarer in a population. The stratified intervention approach is also superior at all levels of intervention effectiveness (Fig. 4b). Both methods' ability to reduce AUROC bias is consistent regardless of the intervention effectiveness.

3.3.2 Probabilistic intervention results

The adjusted posterior risk approach achieves better mitigation of AUROC bias at all levels of outcome prevalence (Fig. 5a). In contrast to the first scenario (Fig. 4a), both correction methods have a better effect in reducing the bias when an outcome occurs more often in a population. The adjusted posterior risk approach is also superior at all levels of intervention effectiveness (Fig. 5b). Both methods' ability to reduce AUROC bias is consistent regardless of the intervention effectiveness.

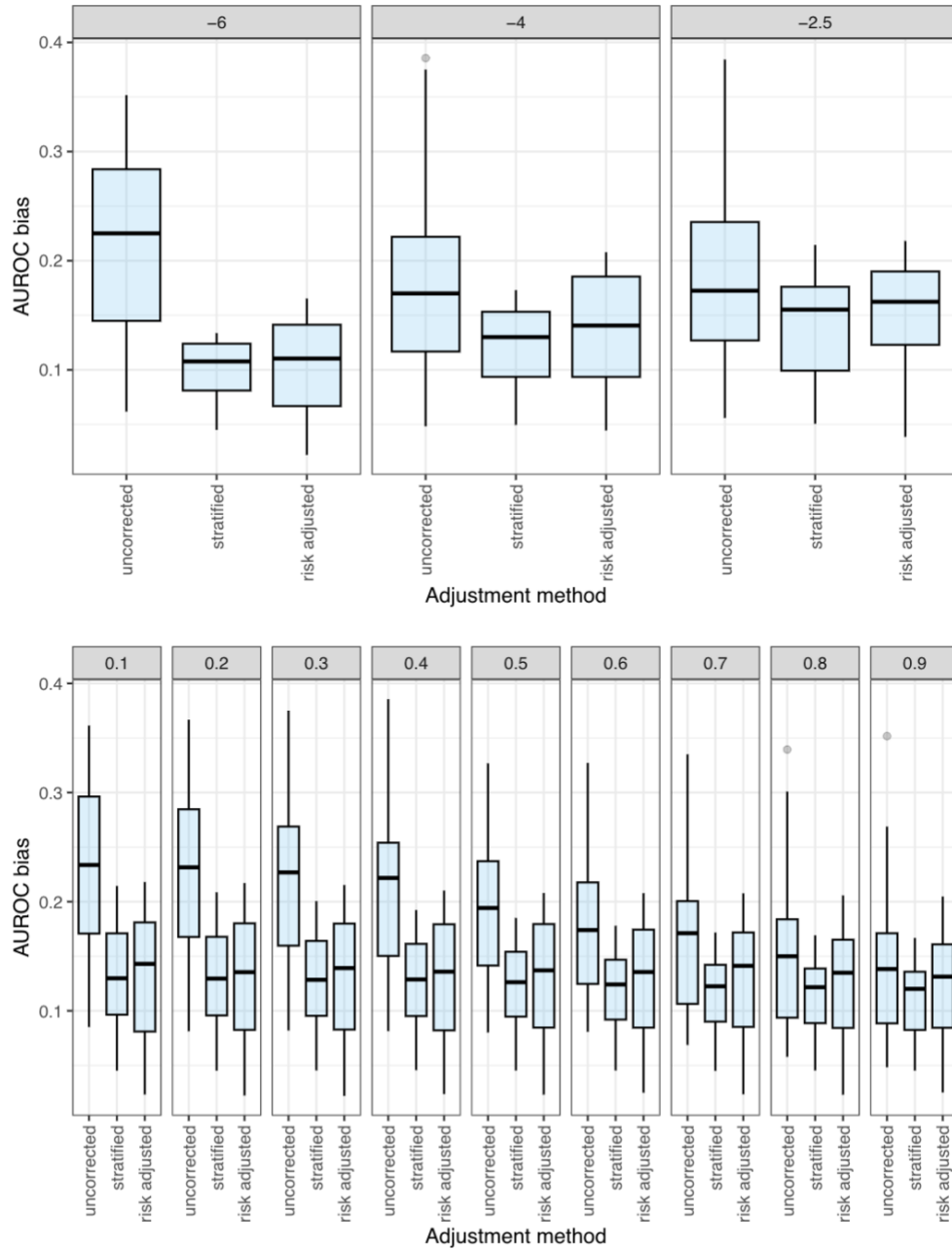


Figure 4: Deterministic intervention: AUROC bias decrease (a) by prevalence; (b) by intervention effectiveness

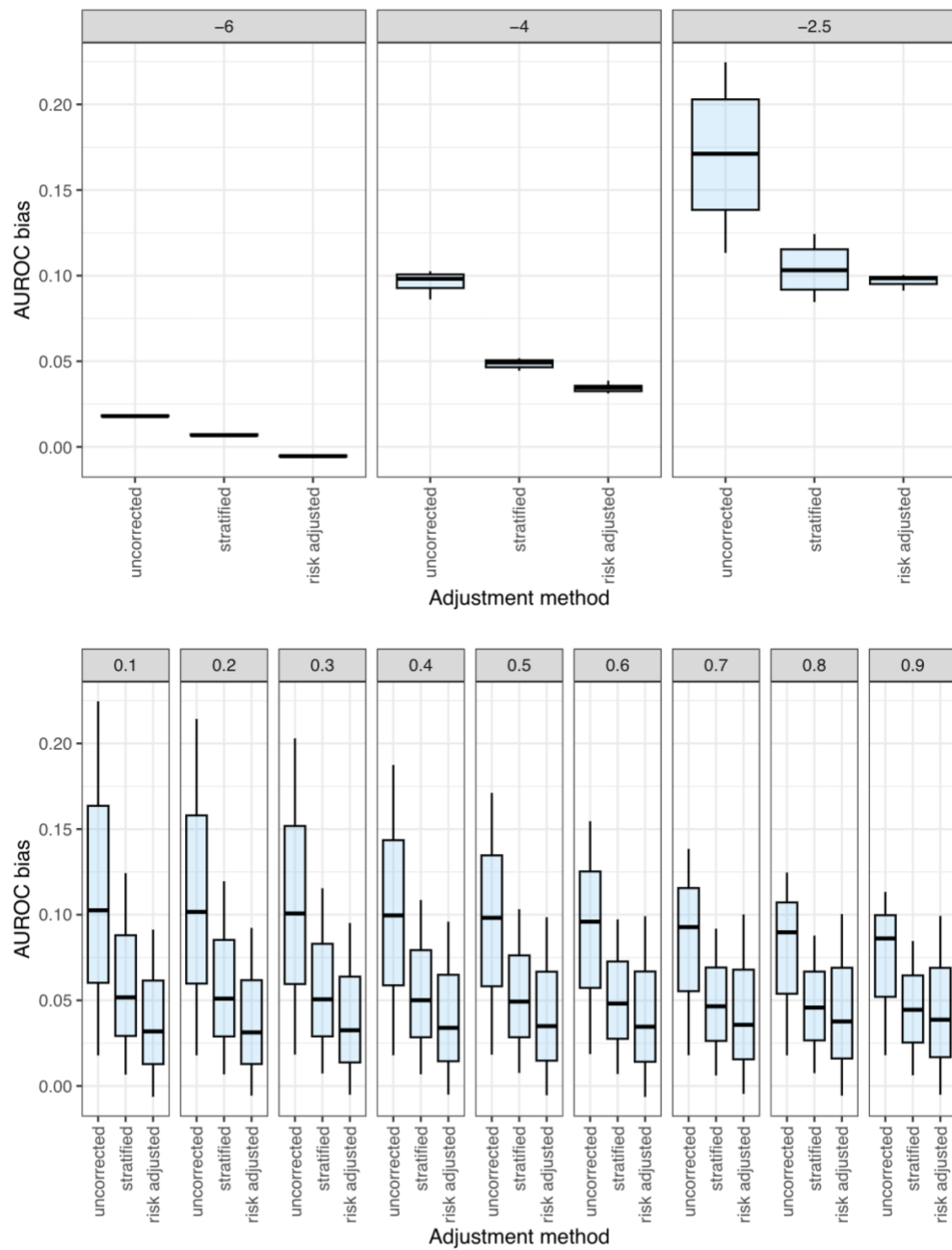


Figure 5: Probabilistic intervention: AUROC bias decrease (a) by prevalence; (b) by intervention effectiveness

4. Discussion

In this study, we considered two intervention scenarios implemented in medical services and illustrated the factors that influence the performance of CPMs: the effectiveness of the intervention, the prevalence of outcome, and the percentage of the population receiving the intervention.

Deterministic intervention scenarios can often be seen in medical admission guidelines published by hospitals (Health and Excellence 2018) and medical insurance companies (CIGNA 2019). There could be an explicit risk score such as the “CURB 65” score for community-acquired pneumonia or a “Blatchford” score that calculates the risk of major Gastrointestinal hemorrhage. This kind of intervention is more common for urgent or major illnesses that require standardized tools for aiding decisions. Predicting patient outcomes without the knowledge of whether they have received intervention is generally more difficult when the outcome is rare, intervention is effective, and when intervention is given to all patients who are at least moderately risky to get the outcome. In this case, the stratified intervention approach effectively reduced the model bias, especially for outcomes with lower prevalence. By separating patients into those who have intervention and those who don’t, we were able to obtain a more accurate estimate within each group. Because of its individualized, needs-based, and least-restrictive philosophy, probabilistic medical intervention is an ideal and comprehensive model of medical care. It is appropriate in contexts where an outcome is commonly seen in the population and medical resources are inexpensive. Predicting patient outcomes without the knowledge of whether they have received intervention is generally more difficult when the outcome is common, and intervention is effective. The commonality of the outcome increases the percentage of people who receive care. Along with strong intervention, many patients would have been

cured already when their outcome is observed. While this flipped outcome reduces a CPM's performance, the adjusted-posterior-risk approach effectively reduced the model bias, especially for outcomes with higher prevalence. Typically, a patient's clinic visits and medication are recorded in EHR data. Incorporating this knowledge of whether patients have received treatment to infer the underlying effectiveness of the intervention is helpful as this piece of additional information helps medical professionals adjust their evaluation of the patient's risk.

While our simulation results shed light on the important role intervention plays in the performance of CPMs as well as proposed effective approaches to improve model performance, there are several limitations. First, there are certain assumptions of our correction methods, particularly the adjusted risk approach. To implement this correction, we assume accurate estimation of intervention effectiveness. This assumption is not necessarily fulfilled, and additional validation is needed. Second, both stratified intervention approach and adjusted posterior risk are based on the knowledge of intervention of individual patient. When a patient visits different healthcare systems that do not share medical records with each other, it is hard to capture an intervention, hence the true outcome can be masked without notice. Third, while racial (Obermeyer et al., 2019) and gender bias (Estiri et al., 2022) have been important topics to investigate in research on CPMs, our simulated data without patient demographic information cannot represent real patient distribution or evaluate differential AUROC bias among various groups of population. Future studies should utilize real-world EHR to test the robustness of our correction approaches and develop frameworks to improve CPM performance in scenario where CMI are partially unknown. In addition, the two types of intervention settings in this study may not be enough to address more complicated care pathways, for example, a scenario where patients having risk below a threshold receives no intervention while those having risk above that

threshold are assigned intervention probabilistically. Future research could also consider this hybrid intervention setting.

In summary, section 1 and 2 of this study illustrated that omitting CMI in prediction models lead to AUROC bias. This bias increases as the intervention becomes more effective and as the percentage of patients receiving intervention comes close to 50%. Section 3 proposed two methods to correct this bias, each improves AUROC differentially under different types of interventions. The stratified intervention approach is more ideal for deterministic intervention, and the adjusted-risk approach is more ideal for probabilistic intervention. However, considering the need to satisfy the assumption of accurate estimation of intervention effectiveness for implementing an adjusted-risk approach, a stratified approach is more widely and safely applicable regardless of intervention settings.

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